Crehul Vlad

vladc12.github.io ← Explore my projects and contributions.

Full-Stack Software Engineer: 4+ years building production systems involving real-time video processing, AI integration, and cloud/edge deployments. Experienced in owning features end-to-end from frontend architecture to backend services and deployment.

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EXPERIENCE

Artificial Intelligence Visual Analytics —*Software Engineer*

09/2021 - PRESENT

- **Architected the development of high-impact full-stack applications** (React, Next.js, Node.js, Python) spanning real-time AI monitoring dashboards and internal data-annotation tools.
- **Engineered a pilot AI-assisted mathematics platform**, featuring a "Human-in-the-Loop" validation workflow that allowed educators to audit and ingest AI-generated question sets into production.
- **Built real-time video inference pipelines** (Python, OpenCV, Kafka) processing multiple concurrent RTSP streams for edge-device deployments in retail and traffic sectors.
- **Developed automated synthetic data pipelines** using Python and Blender, generating high-fidelity 3D datasets to accelerate the training of proprietary machine learning models.
- **Managed end-to-end deployment workflows across cloud infrastructure and edge devices**, configuring networking and remote monitoring solutions for client installations.

Continental Automotive Systems — *Intern Software Developer*

07/2018 - 09/2018

- Developed **embedded software** for an ATV prototype, focusing on real-time data acquisition from electronic sensors.
- Executed **hardware-in-the-loop (HIL) testing for electronic components** to ensure system reliability under stress conditions.

SKILLS

- **Languages:** TypeScript, JavaScript (ES6+), Python, HTML5, CSS3/SASS
- **Frontend:** React, Electron, Context API
- **Backend & Full-Stack:** NodeJS, Flask, MongoDB, NextJS, RESTful APIs, JWT Auth
- **Specialized & Media:** Blender, Kafka, ThreeJS, RTSP
- **Tools & Devops:** Git, Docker, Linux (Edge/Remote Management)

ACCOMPLISHMENTS

Semi-finals: Innovation Labs 2018,

- **Co-engineered an electric self-stabilizing mechatronic prototype** within a multidisciplinary team of four, focusing on hardware-software integration using **Arduino**.
- **Developed the gyroscopic stabilization logic** and closed-loop motor control systems to maintain container equilibrium against multi-axis vehicle motion.
- **Contributed to the successful delivery of a functional prototype**, advancing the project to the competition semi-finals through rigorous technical iteration and testing.

LANGUAGES

- **Romanian**: *Native*
- **English**: *Fluent*

PROJECTS

Learning Platform—React, *TypeScript, MongoDB*

- Architected a **"Human-in-the-Loop" AI education system featuring a LeetCode-style interactive interface** for dynamic mathematics problem-solving.
- Engineered **dual-facing dashboards for real-time student performance tracking and administrative tools for teachers** to manage class-level curriculum analytics.
- Streamlined **AI content ingestion by building an internal validation workflow**, allowing educators to audit and approve pre-generated question sets for production.

Event Manager— *NextJS, TypeScript, MongoDB*

- Delivered a **client-facing analytics dashboard to visualize item-scanning fraud** through complex filterable data tables.
- Implemented a robust **Role-Based Access Control (RBAC) system to manage administrative permissions** and secure sensitive event data.

Caster— *NextJS, TypeScript, Flask*

- Developed an **internal spatial data annotation tool** to define coordinates, directional lines, and Polygons (ROI) via HTML5 Canvas.
- Integrated **Flask-based APIs to handle RTSP stream synchronization** and automated thumbnail generation from live video.
- Enabled precise AI triggering for person-detection events based on **user-defined geometric boundaries**.

Event Watcher & Buddy— *Electron, React, NodeJS, MongoDB*

- Built a **cross-platform desktop application for real-time security alerts**, featuring low-latency playback of flagged fraud events.
- Designed "Event Watcher Buddy" to manage automated software updates, version control, and secure user authentication.
- Implemented a **local-first notification system to alert users** within seconds of an AI-flagged event.

Cradle— *Python, OpenCV, Kafka, Tensorflow, Multiprocessing*

- Architected a **real-time video inference pipeline processing multiple concurrent RTSP streams** for surveillance and people traffic monitoring.
- **Optimized multiprocessing workloads to achieve low-latency ML** inference in edge-computing environments.
- Engineered **high-throughput data payloads** to bridge the gap between backend AI detection and frontend monitoring dashboards.

Quartermaster— *NextJS, TypeScript, threeJS*

- Created a **high-capacity data management system for organizing large-scale datasets** and 3D assets with complex dependencies.
- Implemented **in-browser previews for 3D models (Three.js)**, videos, and images, eliminating the need for local downloads during review.
- Developed a **dynamic file-structure generator that supports batch uploads** and time-limited secure sharing links.

EDUCATION

UTCN, Faculty of Electronics Telecommunications and Technology Information
— Bachelor of Engineering

06/2021

Telecommunication Technologies and System specialization.

BACHELOR THESIS

Tacotron 2 DNN Text-To-Speech Web App— *Flask, Tensorflow*

- Engineered a high-fidelity **TTS system based on the Tacotron 2 deep neural network architecture**, implementing custom training pipelines for both English and Romanian phonetics.
- Optimized model performance through specialized datasets, utilizing the **LJSpeech Dataset for English and the Mara Corpus for Romanian** to achieve natural-sounding speech synthesis.
- Developed a **Flask-based web interface to serve real-time inference**, providing a seamless demonstration of the model's linguistic capabilities and neural processing.